

BM4 Fourth-Order Composition

Direct-adjoint splitting and all projection configurations

1 Common composition

Let χ_h be a first-order map and $\chi_h^* = \chi_{-h}^{-1}$ its adjoint. All BM4 configurations use the palindromic twelve-stage composition

$$\Psi_h = \chi_{b_{12}h} \circ \chi_{b_{11}h}^* \circ \cdots \circ \chi_{b_2h} \circ \chi_{b_1h}^*,$$

with alternating adjoint and direct maps in chronological execution order and

$$(b_1, \dots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1),$$

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

The half-cycle coefficients sum to $1/2$. Palindromy makes the composition self-adjoint and its order conditions give designed global order four when the direct-adjoint pair satisfies the assumed smoothness and consistency. The negative a_4 is an intentional backward subflow.

2 Public configurations

The family exposes six classes because projection placement changes the numerical map:

Class	projection placement	nonlinear unknown	un-
BM4Composition	formulation-owned output extraction	none	
ProjectedBM4Composition	arithmetic re-embedding after every stage	none	
MidpointBM4	arithmetic re-embedding after the full cycle	none	
BM4Implicit1	one symmetric projection around the full cycle	$\mu \in \mathbb{R}^{2N}$	
BM4Implicit2	the same projection in simultaneous form	$(Y, \mu) \in \mathbb{R}^{6N}$	
BM4_implicit2	full-diagonal projection of duplicated Z	$\mu \in \mathbb{R}^4$	

The underscore in `BM4_implicit2` distinguishes the fully extended method from the simultaneous physical method `BM4Implicit2`.

3 Doubled physical state

For a physical state $z \in \mathbb{R}^m$, define

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \quad P = \frac{1}{2} \begin{pmatrix} I & I \end{pmatrix}, \quad G = \begin{pmatrix} I & -I \end{pmatrix}, \quad N = G^T.$$

The unprojected composition retains two copies. Stage projection applies EP after each of the twelve factors; midpoint projection applies it once after the complete cycle. These placements must not be conflated because projection and the stage maps do not commute.

4 Implicit physical projection

Let $\Psi_{h,t}$ denote the complete coupled BM4 cycle. The reduced method solves

$$M(\mu) = \Psi_{h,t}(Ez + N\mu), \quad (1)$$

$$r(\mu) = GM(\mu) + 2\mu = 0, \quad (2)$$

$$z^+ = P(M(\mu) + N\mu). \quad (3)$$

With $J_{\text{BM4}} = D\Psi_{h,t}(Ez + N\mu)$, the exact Newton matrix is

$$D_\mu r = GJ_{\text{BM4}}N + 2I.$$

The simultaneous method solves

$$F(Y, \mu) = \begin{pmatrix} Y - N\mu - \Psi_{h,t}(Ez + N\mu) \\ GY \end{pmatrix} = 0,$$

whose block Jacobian is

$$\begin{pmatrix} I & -(I + J_{\text{BM4}})N \\ G & 0 \end{pmatrix}.$$

Its Schur complement is the reduced Newton equation, proving equivalence of the two ideal roots. Both formulations accept exact Newton or good Broyden; this solver choice changes work and finite-tolerance behavior, not the defining projected map.

5 Fully extended configuration

For one non-autonomous guiding-centre particle introduce

$$Z = (x, y, t, k), \quad K(Z) = H(t, x, y) + k.$$

`BM4_implicit2` duplicates Z to an eight-dimensional splitting state, executes the same twelve-stage coefficient schedule, and projects the full diagonal with a four-component multiplier. The accepted physical solution contains (x, y) while diagnostics retain t, k , and the drift of K . This configuration is a genuine extended-state map and is not algebraic workspace for `BM4Implicit2`.

6 Geometric interpretation and finite solves

For an exact adjoint pair, the base composition is symmetric and fourth order. The symmetric implicit projection is designed to preserve the inherited geometric structure at its exact root. Arithmetic stage or cycle projections define separate maps and must be assessed directly. Finite Newton or Broyden tolerances perturb every ideal statement; the implementation therefore records nonlinear work and exposes step maps for Jacobian-based audits.

7 Implementation correspondence

The shared composition is in the BM4 module's `_core.py`; the midpoint, reduced, simultaneous, and fully extended variants are in its neighboring modules. Two detailed derivations accompany this theory:

- `implicit-reduced.tex`; and
- `implicit-simultaneous.tex`.

The runtime architecture lives under `../simulation/`; shared physical dynamics are documented outside the model tree.